Why Study Money, Banking, and Financial Markets?

Chapter 1

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview

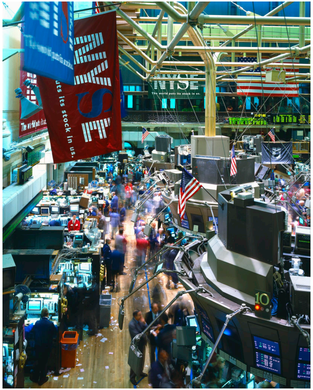
- How the workings of **financial markets** such as bond, stock and foreign exchange markets affect your everyday life
- How **financial institutions** such as banks, investment and insurance companies work
- The **role of money** in the economy

Finance determines **where money goes**, and therefore how the economy grows.

Outline

Why Study Financial Markets?

- Why study them?



The trading floor of the New York Stock Exchange.

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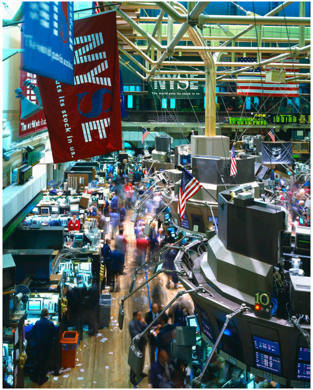
Why Study Financial Markets?

- Why study them?
  - **High importance**

Finance raises **efficiency**: it does not directly create output, but decides which projects get funded.



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Why Study Financial Markets?

- Why study them?
  - **High importance**
  - **High complexity**

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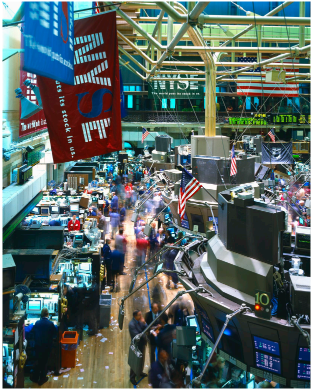

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High Importance of Financial Markets

- Financial markets determine:
  - who receives funding
  - at what cost
  - for how long
  - and how risks are shared
- Therefore, financial markets directly affect:
  - Economic growth: whether capital flows to high-return projects
  - Innovation: whether risky R&D can obtain long-term equity and venture funding
  - Macroeconomic stability: credit booms and busts amplify business cycles
  - Policy transmission: monetary policy works through financial institutions

Finance determines where money goes, and therefore how the economy grows.

A Thought Question

- Can you think of two countries with:
  - similar culture and historical background,
  - but very different long-run economic performance?

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  - Human capital?
  - Institutions?

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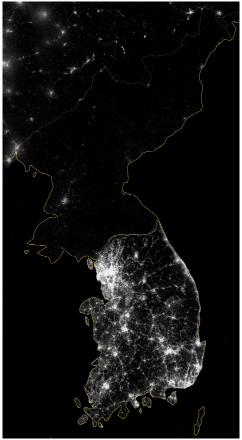
  - Human capital?
  - Institutions?
  - **Financial systems?**

A Real Example: South Korea vs. North Korea

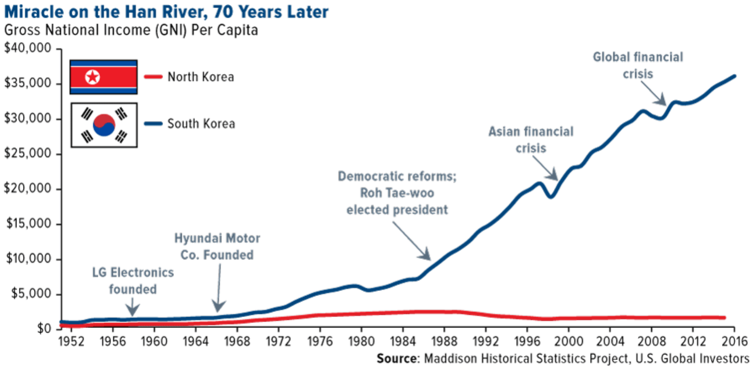
- South Korea and North Korea share:
  - common history, language, and culture
  - similar initial levels of development after the 1950s
- But today:
  - South Korea is a high-income, innovation-driven economy
  - North Korea remains extremely poor and technologically backward



The Korean peninsula at night, seen from space (2012).

Wikimedia Commons, NASA, Public domain

South Korea vs. North Korea: GDP per Capita



Real GDP per capita in South and North Korea since the 1950s.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Financial Repression vs. Financial Deepening

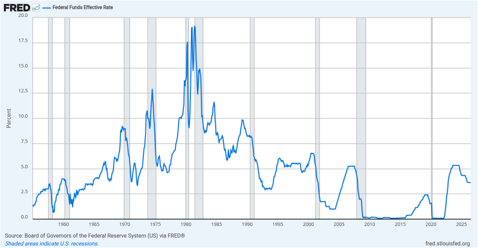
- **Financial Repression**
  - Government controls interest rates and credit allocation
  - Capital flows and financial institutions are restricted
  - Capital allocation is politically or administratively driven
- **Financial Deepening**
  - Financial markets expand in size and diversity
  - Interest rates and risk are market-determined
  - Capital allocation becomes more efficient and market-based

A Thought Question: The Fed Raises Rates

The Federal Reserve is raising the federal funds rate by $\frac{1}{2}$ of a percentage point. What potential effects might this have on the following?

1. the interest rate on an automobile loan
2. the price of houses
3. the difficulty of securing a job next year

Hint: Consider how higher interest rates influence borrowing costs, investment decisions, and overall economic activity.



Effective federal funds rate since 1955; shaded bars are recessions.

Source: FRED, Federal Reserve Bank of St. Louis

The Importance of Financial Markets

- Financial markets and institutions not only influence your **everyday life**, but also shape the flow of **trillions of dollars** throughout the economy
- This, in turn, affects:
  - **Business profits**
  - **Production of goods and services**
  - **Economic well-being** of countries globally

The financial crises of **2007–2009**, triggered by the collapse of housing markets in several countries, and the **recent pandemic** have demonstrated the **deep and far-reaching consequences** of financial market disruptions on economic agents worldwide.

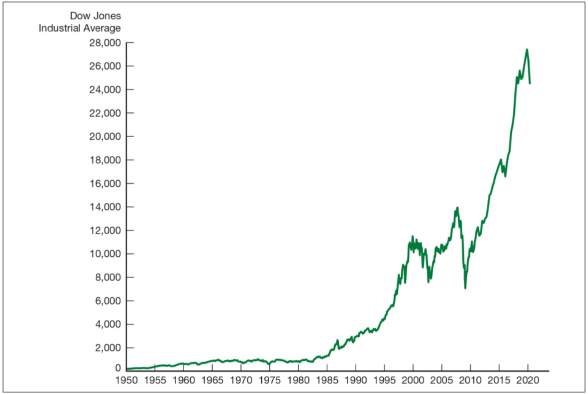
The Bond Market and Interest Rates

- A **security** (financial instrument) is a claim on the issuer's future income or assets.
  e.g., stocks or mortgage-backed securities
- A **bond** is a debt security that promises to make payments periodically for a specified period of time.
  e.g., U.S. Treasury bonds, corporate bonds, or municipal bonds
- An **interest rate** is the cost of borrowing or the price paid for the rental of funds.

The bond market is where interest rates are determined – and interest rates are the prices that matter most for the rest of this course.

Stock Prices

- How can we explain the 2008 stock market crash?
- What factors drive stock price fluctuations?
- Why does the government encourage stock market growth?



Stock prices as measured by the Dow Jones Industrial Average, 1950–2020.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Pricing Bonds by Quoted Coupon Rate

Two bonds with similar risk

- **Bond A**
  - Face value: 1,000 Coupon: 5% (annual)
  - Maturity: 5 years Price: 985
- **Bond B**
  - Face value: 1,000 Coupon: 7% (annual)
  - Maturity: 6 years Price: 1,090

Quick decision?

- A is **cheaper**, but pays **lower** interest
- B is **more expensive**, but pays **higher** interest

One complication

- Different maturities (5 vs. 6 years)

Switching the Quote: Yield to Maturity (YTM)

Same two bonds (similar risk)

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Quote using yields (YTM)

- **Bond A: 5.35%**
- **Bond B: 5.21%**

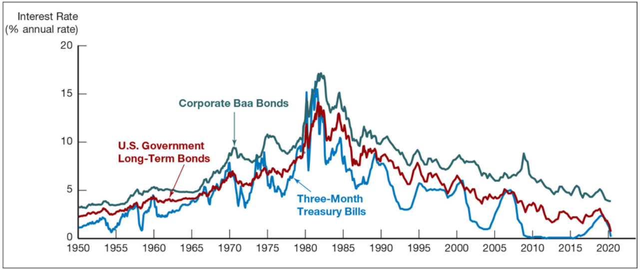
Decision (if risk is similar)

- Higher yield $\Rightarrow$ better return
- Choose **Bond A**

YTM summarizes the bond's return by converting all cash flows into a single rate.

Interest Rates on Selected Bonds

- Why are interest rates difficult to analyze?
- Which interest rates do investors care about?
- How do we explain differences across rates?



Interest rates on selected bonds, 1950–2020.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Why Study Financial Institutions and Banking?

1. **Why do banks exist?**
   - What limitations of bonds and stocks do banks overcome?
2. **How do banks differ from other financial institutions?**
   - What defines a bank?
   - Are there hybrid or transitional institutions?
   - Can fintech replace traditional banks?

These questions motivate our study of financial intermediation and financial innovation.

Financial Intermediaries and Banks

Financial Intermediaries

Institutions that **borrow funds** from people who have saved and in turn **lend funds** to those who need them.
Examples: banks, insurance companies, finance companies, pension funds, mutual funds, and investment companies.

Banks

Banks accept deposits and make loans, playing a central role in:

- capital allocation
- risk management
- monetary policy transmission

Financial Innovation and Financial Crises

Financial Innovation

The development of **new financial products and services** that:

- improve efficiency in the financial system
- expand access to finance
- reshape financial intermediation

E-Finance: the electronic delivery of financial services.

Financial Crises

Major disruptions in financial markets characterized by:

- sharp declines in asset prices
- failures of financial and nonfinancial firms

Outline

Why Study Money and Monetary Policy?

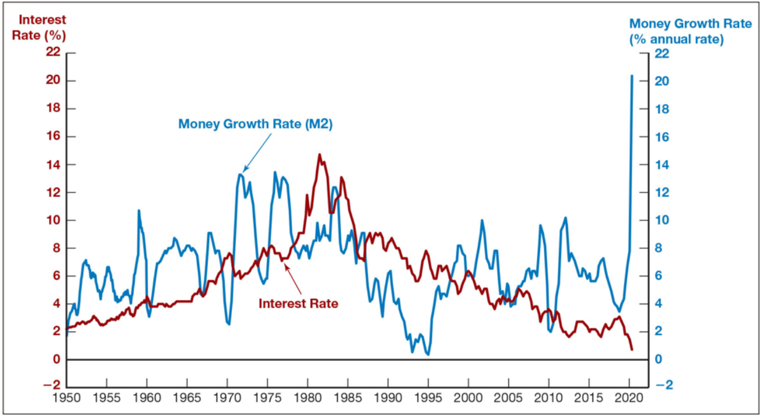
The Role of Money: Money, defined as anything generally accepted as payment for goods, services, or debt repayment, plays a key role in driving business cycles.

Impact of Business Cycles: Recessions (unemployment) and expansions affect everyone in the economy.

Monetary Theory: Links changes in the money supply to:

- Aggregate economic activity
- The price level

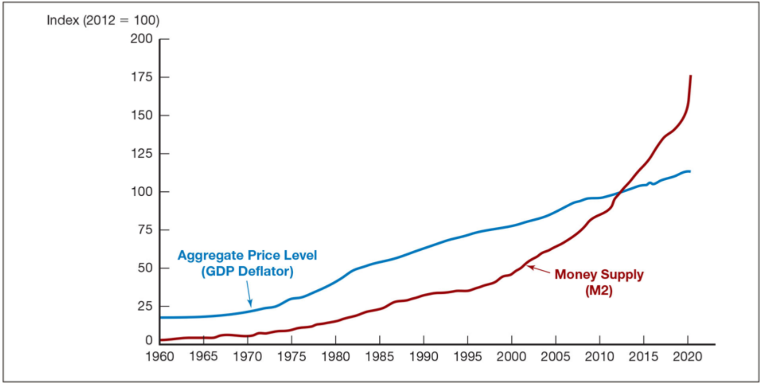
Money Growth and the Business Cycle



Money growth (M2 annual rate) and the business cycle in the United States, 1950–2020.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

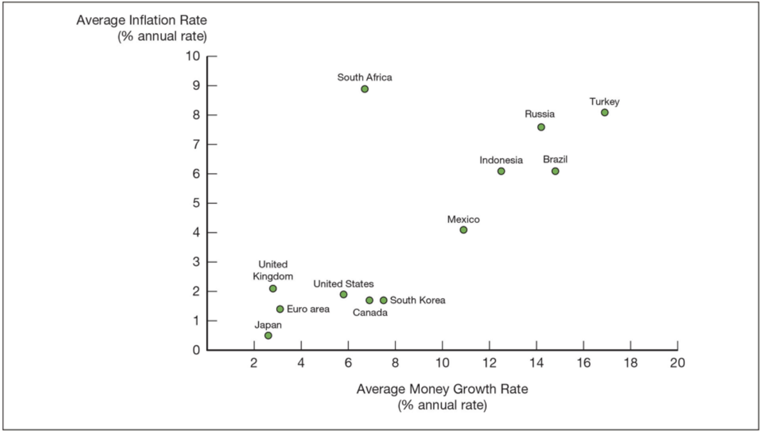
Aggregate Price Level and the Money Supply



Aggregate price level and the money supply in the United States, 1960–2020.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Inflation Versus Money Growth Across Countries



Average inflation rate versus average rate of money growth for selected countries, 2009–2019.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Money and Interest Rates

What Are Interest Rates? Interest rates represent the **price of money**.

Historical Relationship:

- Before 1980: the rate of money growth and long-term Treasury bond interest rates were closely linked
- After 1980: the relationship became less clear, but money growth remains an important factor influencing interest rates

> Money growth, inflation, and interest rates move together over long periods – but the link is looser than it used to be.

Fiscal Policy and Monetary Policy

Monetary Policy: The management of the **money supply** and **interest rates**, conducted in the United States by the **Federal Reserve System (Fed)**.

Fiscal Policy:

- Focuses on **government spending** and **taxation**
- **Budget Deficit:** expenditures exceed revenues in a given year; must be financed by borrowing
- **Budget Surplus:** revenues exceed expenditures in a given year

Is Monetary Policy Easy to Implement?

Fiscal policy interactions

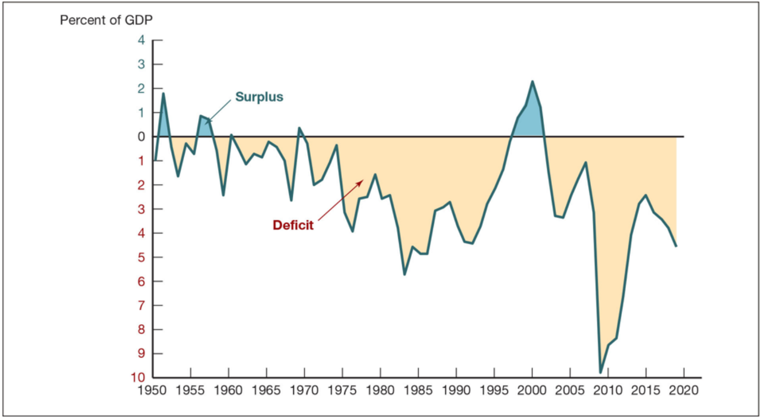
- How government deficits affect the money supply

Complexity of implementation

- What are the objectives?
- What policy tools are available?
- What difficulties arise in practice?
- How does the central bank adjust policy continuously?

The Marriner S. Eccles Federal Reserve Board Building, Washington, D.C.

Government Budget Surplus or Deficit



Government budget surplus or deficit as a percentage of GDP, 1950–2019.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Outline

The importance of financial markets

Money, monetary policy, and the business cycle

The importance of international finance

Appendix

The Foreign Exchange Market

Foreign exchange market: the market where **funds are converted** from one currency to another.

Foreign exchange rate: the price of one currency in terms of another currency.

Role of the market: the foreign exchange market determines the foreign exchange rate.

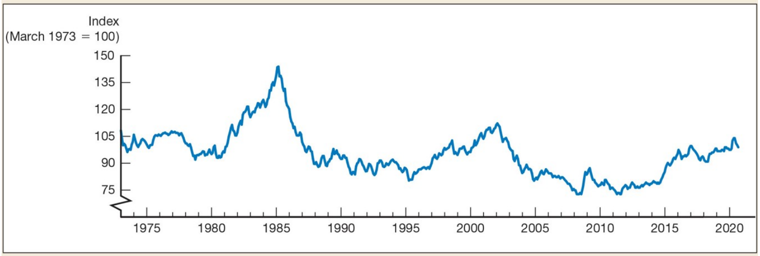
Why Study International Finance?

Global Integration: Financial markets are becoming increasingly integrated worldwide.

Impact of the International Financial System: The international financial system significantly influences domestic economies. Key questions include:

- How does a country's choice of exchange rate policy affect its monetary policy?
- How do capital controls influence domestic financial systems and overall economic performance?
- What role should international financial institutions, such as the **IMF**, play in the global economy?

Exchange Rate of the U.S. Dollar



Exchange rate of the U.S. dollar, 1973–2020.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Chapter Summary

- Financial markets decide who gets funding, at what cost, and how risk is shared – so they shape growth, innovation, and stability
- Bonds and interest rates, stock prices, and financial intermediaries (especially banks) are the core objects of study
- Money growth is linked to the business cycle, the price level, and interest rates; the Fed manages it through monetary policy
- Fiscal policy (spending, taxes, deficits) interacts with monetary policy
- Exchange rates and the international financial system increasingly influence the domestic economy

Outline

Total Output = Total Income

Total Output (GDP)

- The market value of all **final goods and services** produced within a country in a given period
- Final goods and services only (exclude intermediate goods)
- Financial transactions are excluded
- Sales of second-hand goods are excluded

Total Income

- The total income earned by production factors in the process of producing goods and services
- Production factors: land, capital, labor, and entrepreneurial ability
- Corresponding incomes: rent, interest, wages, and profit

Real vs. Nominal Variables

- **Why distinguish real and nominal variables?**
- **Economic growth**
  - Increase in real output of final goods and services
  - Not just higher prices
- **Interest rates**
  - Is a 50% nominal interest rate necessarily high?
  - Inflation must be considered
- **Goal:** eliminate the effect of price changes

The Price Level

- How do we measure overall prices?
- Why do we need stock indexes?
  - Individual stock movements differ
  - Index uses an average
- Why do we need price indexes?
  - Individual prices fluctuate differently
  - Index summarizes overall price movements
- Price indexes depend on:
  - Which goods and services are included
  - Who the index is designed for

Measuring the Price Level

- Common price indexes:
  - GDP Deflator
  - Consumer Price Index (CPI)
  - Producer Price Index (PPI)
- Differences:
  - Goods and services included
  - Weights assigned
  - Target population

> A price index summarizes the overall movement of many individual prices in a single number – which goods are included and how they are weighted determines what it measures.

How to Calculate Growth Rates

- Standard formula:

$$\text{Growth rate of } x = \frac{x_t - x_{t-1}}{x_{t-1}}$$

- Limitation:
  - Increases and decreases are not symmetric
- Example:
  - Price rises by 10%, then falls by 10%
  - Final value: $P(1.1)(0.9) = 0.99P$